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Estimating Earthquake Early Warning Effectiveness via Blind Zone Sizes: A Case Study of the New Seismic Network in Chinese Mainland

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Using a theoretical network-based method along with Monte Carlo random sampling with 50,000 iterations, we predicted the spatial distribution of the blind zone radii and quantified the warning effectiveness of the new EEW system in Chinese mainland based on its network configuration. Additionally, we investigated the impact of “soft” blind zones caused by potential station failures on the performance of the new EEW system in China. This file serves as a valuable complement to the main text, featuring earthquake plots, previous and new station distributions across Chinese mainland, as well as maps illustrating inter-station distances for both the previous and new networks. We also provide estimates of the mean and standard deviation of blind zone radii, based on scenarios with a minimum of four and five trigger stations, as well as their changes in mean blind zone radii and lead time compared to scenarios assuming three trigger stations.

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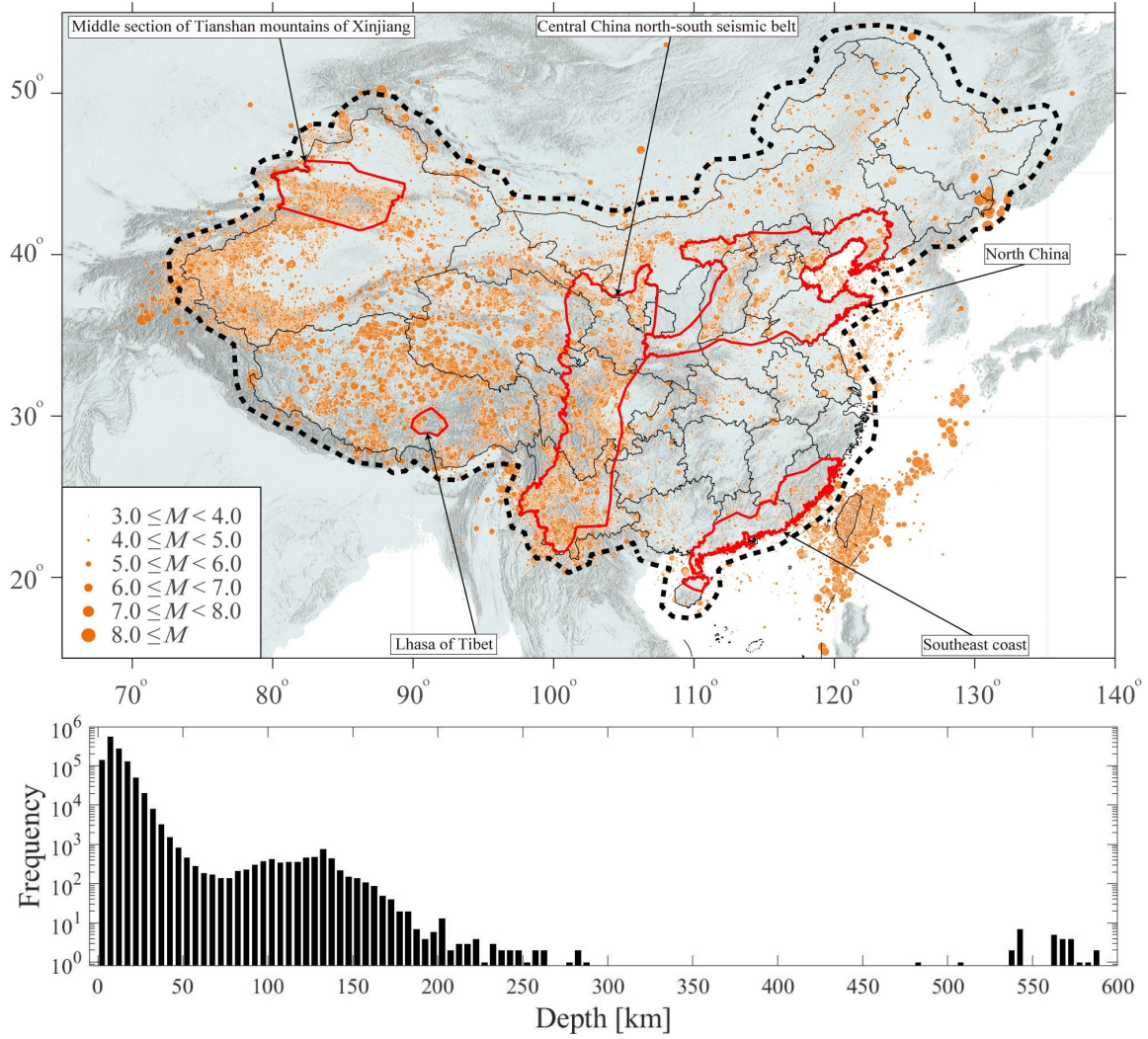


Figure S1. (Top) The epicenters (from China Earthquake Networks Center, CENC) of approximately 92,000 earthquakes with $M_L \geq 3.0$ that occurred from 01/01/1970 to 29/06/2022 in China and its adjacent regions. The areas delineated by the red solid lines represent the five key earthquake early warning (EEW) regions developed by the National System for Fast Report of Intensities and Earthquake Early Warning Project of China, which are expected to be operational after 2024. The thin gray lines indicate the provincial administrative boundaries of China. The region defined as "inside" in this study corresponds to the area within 100 km outside of Chinese mainland's boundary (black bold dashed line), based on the coverage of the seismic stations of the CENC. (Bottom) Depth-frequency distribution with statistical bins of 5 km for earthquakes from 01/01/1970 to 20/11/2024 in China and its adjacent regions.

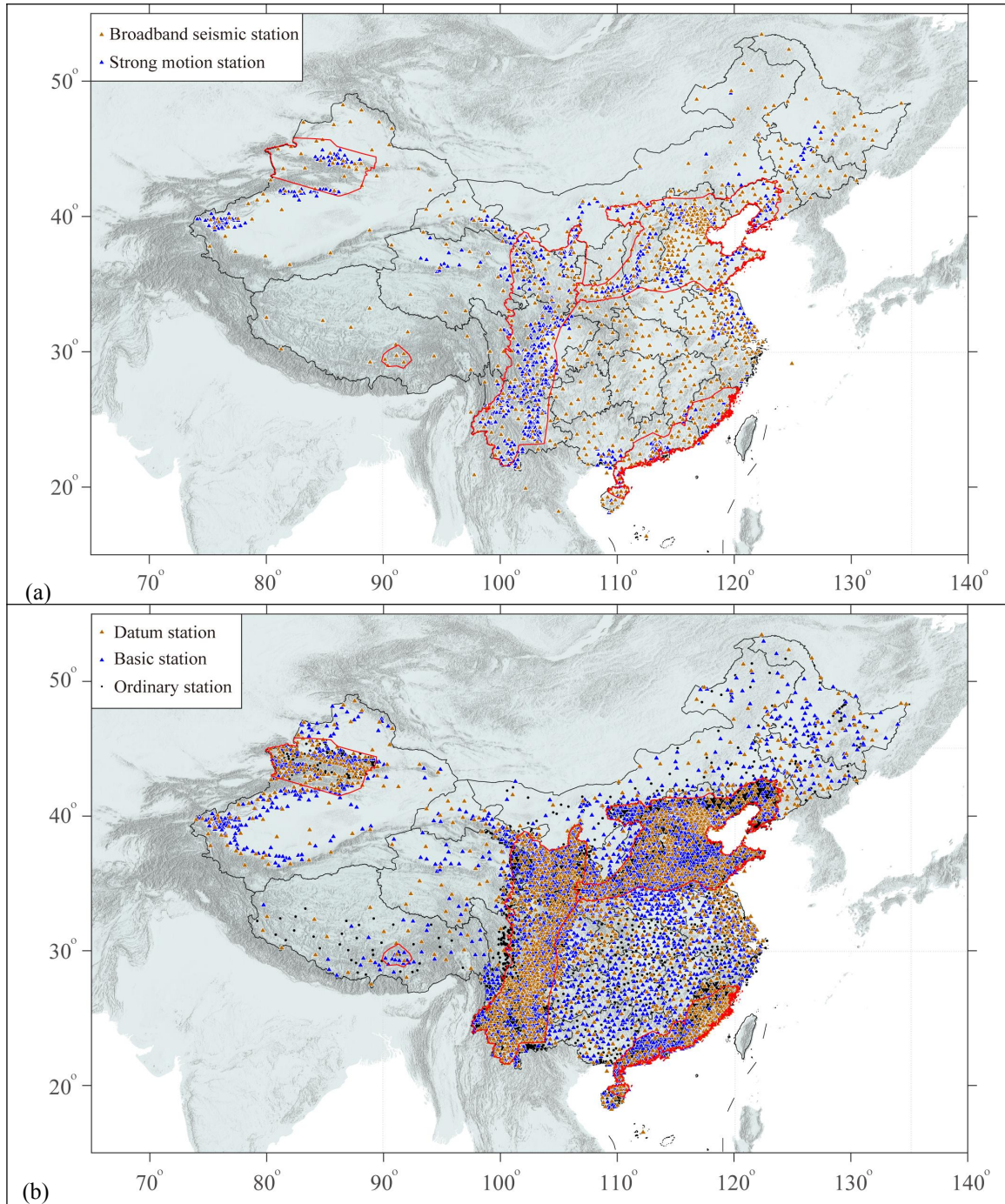


Figure S2. (a) Previous (as of 2018) stations as part of the CENC and the China Strong Motion Network Center (CSMNC). (b) New (as of 2023) stations to be newly deployed or upgraded in the National System for Fast Report of Intensities and Earthquake Early Warning of China. The areas delineated by the red solid lines represent the five key EEW regions.

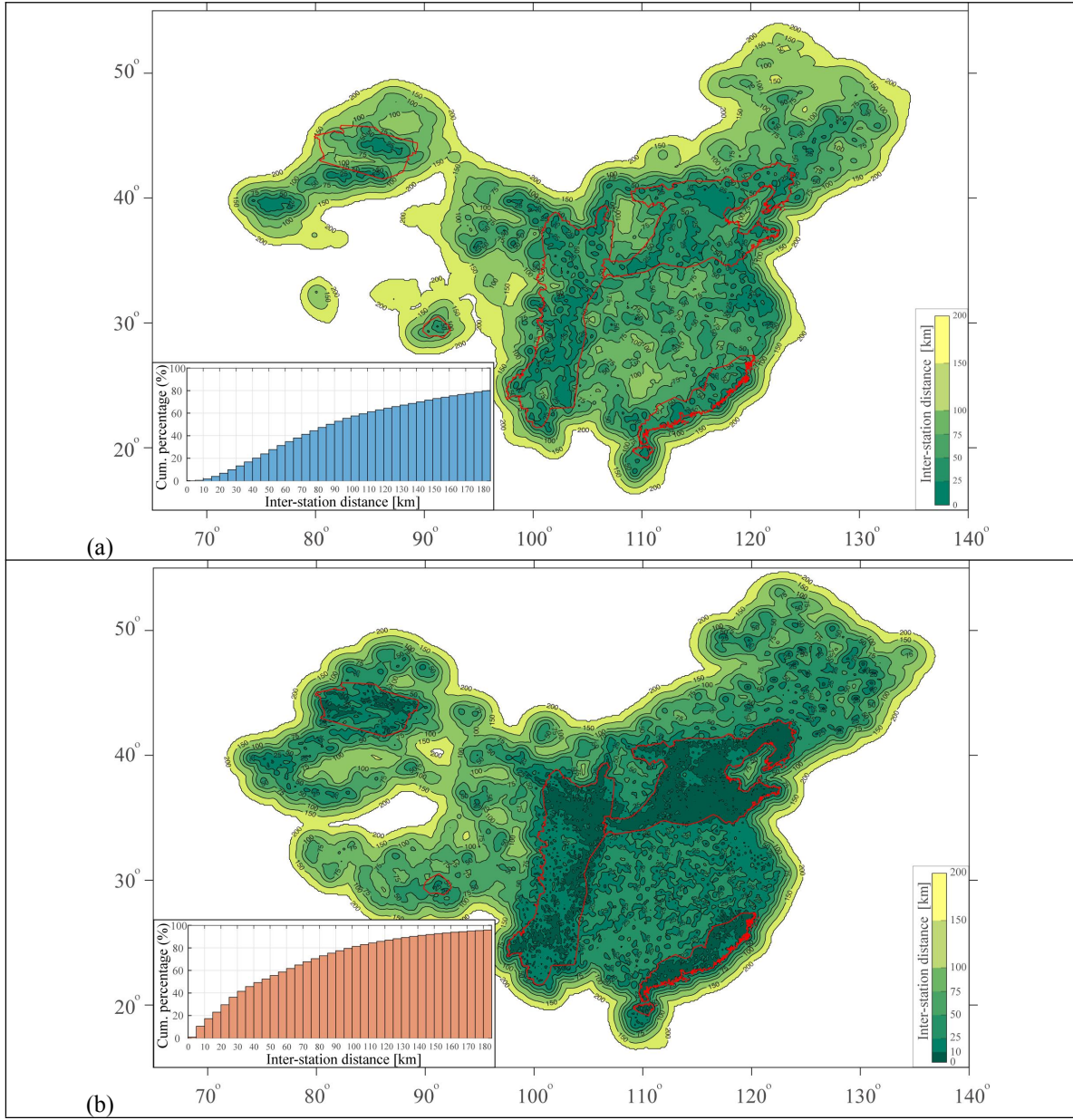


Figure S3. Maps showing the inter-station distances for the (a) previous and (b) new networks. For any given site (assuming a grid of $0.1^\circ \times 0.1^\circ$ resolution), the inter-station distance have been calculated as the average distance to the three closest stations (Kuyuk and Allen 2013; Li et al. 2016; Li et al. 2021). The histograms in the bottom left depict the cumulative percentage distribution of the inter-station distances with statistical bins of 5 km inside Chinese mainland. The areas delineated by the red solid lines represent the five key EEW regions.

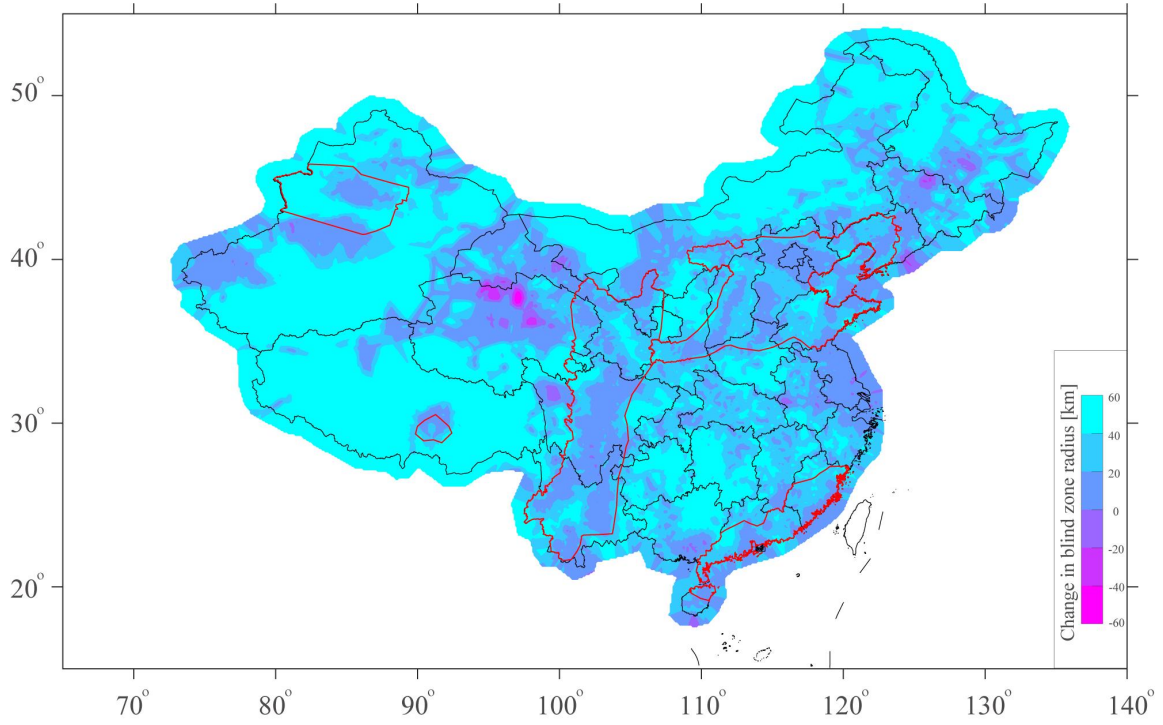


Figure S4. Map showing changes in mean blind zone radii between the previous and new networks with a minimum of three trigger stations (Figures 2a and 3a). Positive values indicate relative reduction in mean blind zone radii of the new network compared to the previous network, while negative values indicate expansion of mean blind zone radii of the new network relative to the previous network. Limited to inside regions of Chinese mainland, encompassing both comprehensive coverage of the previous and new networks.

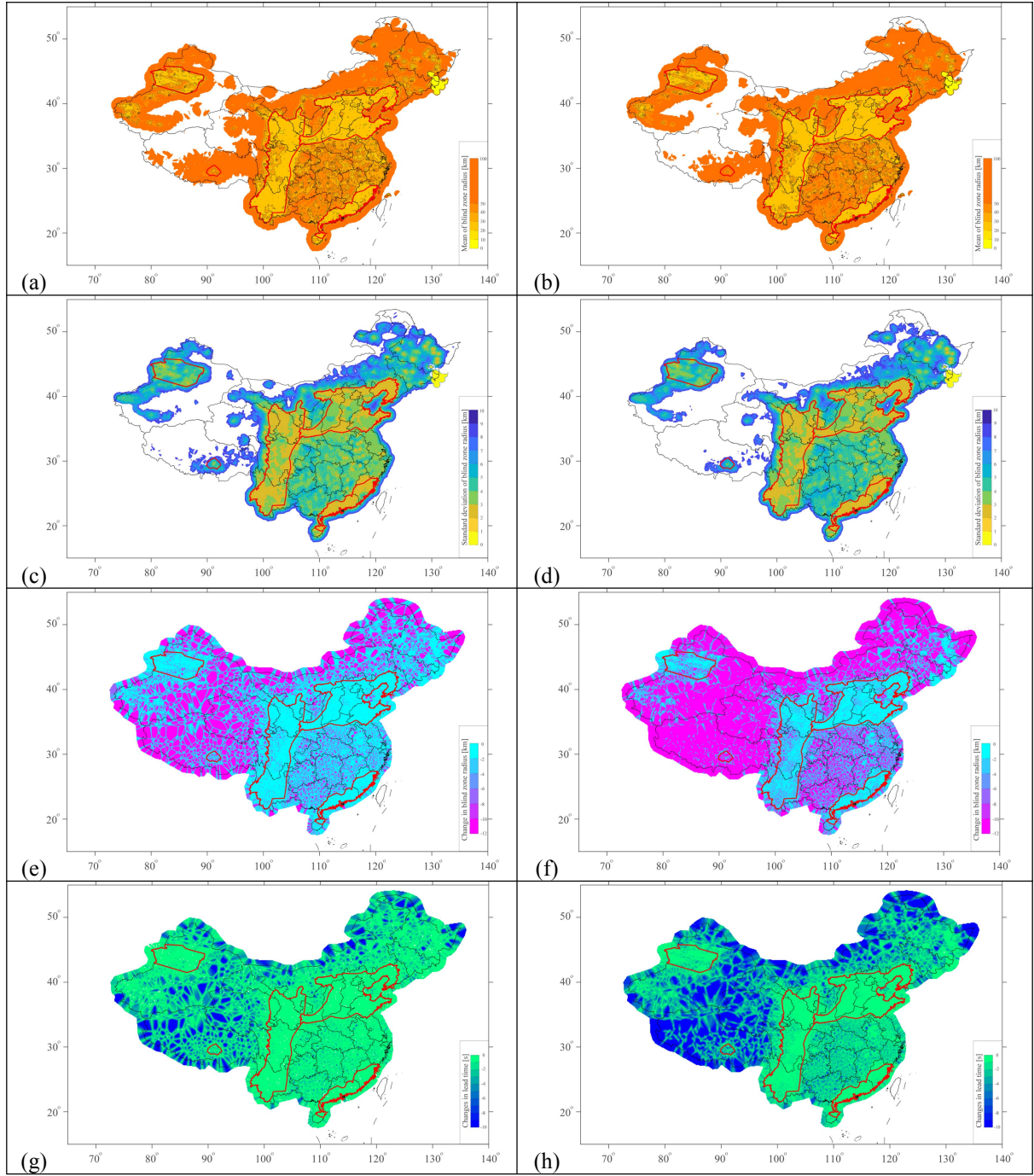


Figure S5. Each row of maps respectively shows: the mean of the estimated blind zone radii, the standard deviation of the estimated blind zone radii, changes in mean blind zone radii, and the lead time (available warning time) changes due to changes in mean blind zone radii for the new network, compared to a minimum of three trigger stations. The left column and right column correspond to scenarios assuming a minimum number of four and five trigger stations, respectively. The curves of the cumulative distribution for (a) to (d) inside Chinese mainland can be found in Figure 6 in the main text.

Reference

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- Li JW, Böse M, Feng Y, Yang C (2021) Real-time characterization of finite rupture and its implication for earthquake early warning: Application of FinDer to existing and planned stations in Southwest China. *Front Earth Sci* 9:699560.